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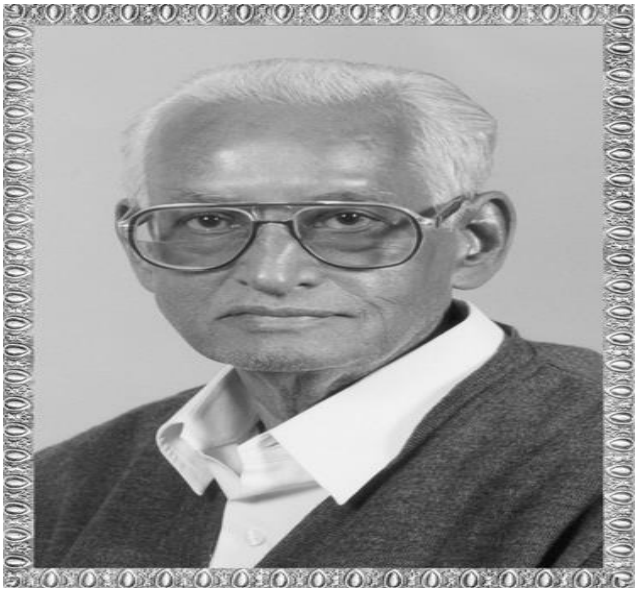

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AEC- CS-26: FUNDAMENTALS OF IOT

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UNIT-1 INTRODUCTION OF IOT

INDEX

TOPICS
Introduction to the Internet of Things (IoT)
History and Evolution of IoT
Key Concepts and Definitions
Applications and Use Cases of IoT
Challenges and Opportunities in IoT

Q-1 What is Iot? Introduction of iot.

Ans:-

- The **Internet of Things (IoT)** describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- The term 'Internet of Things' was found in **1999** by the computer scientist Kevin Ashton. Who was British technology pioneer.

Q-2 Explain history and evaluation of IOT.

- The **Internet of Things (IoT)** describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- Here some overview of history and evaluation of iot
  - **Early beginnings (1960 to 1980):-**

    1) First network device:-

    ➔ The first network device were developed in the year 1960 with creation of first network **ARPANET.**

    2) Machine to machine communication(M 2 M):-

    → in the 1980 m2m communication emerged ,enable device to communicate with each other without human intervention.

  - **The emerged of IOT:-**

    ➔ The term 'Internet of Things' was found in **1999** by the computer scientist Kevin Ashton. Who was British technology pioneer.

    ➔ Who work in procter and gamble at that time , used the term to describe the network of physical object that could be connected to internet.

    ➔ Other key figures who have contributed to the development of iot include peter T. Lewis , Vint Cerf,.

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- ➔ Iot is network of physical devices likes vehicles , home appliance , other item embedded with sensor, software and connectivity allowing them to collect and exchange data.
- ➔ Iot created more connected and automated world.
- ➔ The goal of iot is make life easier.
- ➔ Example of iot devices are smart home, security camera, lock system, lights, smart watches , industrial sensor like pressure checker and vibration checker, medical devices like pacemakers and pump.
- ➔ Iot used in many sector like health care, industrial automation , smart cities , transportation , home automation.

- ➔ Wireless sensor networks (wsns) is emerged in the early 2000 that enable device to communicate each with wireless.

- **IOT Expansion and usages:-**

  1) **Smart home and cities:-**

  ➔ The concept of smart home and cities emerged with integration of IOT device into daily life.

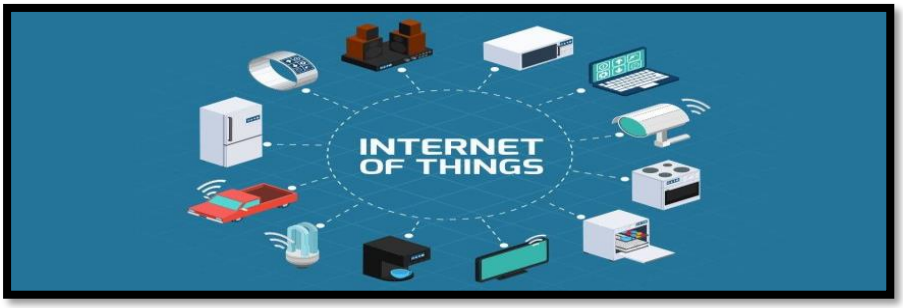
  2) **wearable technology:-**

  → wearable technology such as fitness tracker and smart watches become popular.

  3) **Industry 4.0:-**

  → the concept of industry 4.0 emerged with the integration of iot , ai and robotics into manufacturing.

- Iot also used in AI, ML, 5g Network security and privacy concern.



Q-2 Explain iot definition and key concept of IOt.

- The **Internet of Things (IoT)** describes the network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.
- The most important features of IoT on which it works are connectivity, analyzing, integrating, active engagement, and many more
  ➔ **key concept of iot:-**
- Here are some key concept of IOT are
  ➔ data collection ,
  ➔ connectivity,
  ➔ sensor and actuators ,
  ➔ processing and analysis,
  ➔ user interface,
  ➔ machine to machine communication,
  ➔ security ,
  ➔ protocol ,
  ➔ ai and machine learning,
  ➔ autonomy.

1) **Date collection:-**

- Iot collect data using many ways like it collect data using sensor , actuators , api or machine learning.

2) **Connectivity:-**

➔ iot device can connect to the internet , communicate with each other and exchange data.

➔ IoT devices can communicate with each other and with the cloud over the internet.

➔ they connect to an IoT gateway, which acts as a central hub for sending and receiving data.

3) **Sensor and actuators:**

- Iot devices are used with sensor.
- It collect data and pass and that perform action based on data.
- Iot pass data using actuators. it is one type of machine that receive input signal such as electrical , air pressure , water pressure and more.

4) **processing and analysis:**

- Iot device generate much amount of data which is analysed to gain insights and make decision and improve process.
- It first take data and then done whole analysis and then take decision or give result.

5) **User interface:-**

- A user interface (UI) in the Internet of Things (IoT) is the way that users interact with and understand the data captured by IoT devices.
- The goal of a well-designed UI is to make the user experience easy so that users can achieve their desired outcome with minimum effort.

6) **Machie to machine communication:-**

- Iot device can communication with each other by machine
- Iot device communicate with each other without human intervention.

7) **Security:-**

- Iot device require proper security to measure to protect against cyber threats and data.
- Security is main require because today world more and more people and industries use this concept so highly security concept require.

8) **Protocol:**

- Iot is one type of network so it require protocol system.
- Internet of Things (IoT) protocols are a set of rules and standards that allow devices to communicate with each other and with the internet.

9) **AI and Machine learning:-**

- Iot device can use ai and ml to analyse data , make prediction and improve performane.

10) **Autnomy:-**

- Iot device can operate automatic ,make decision and taking action without help of human.

Q-3 Explain Application of iot. Or explain use cases of iot

Ans:-

- The Internet of Things (IoT) has many applications in both the private and public sectors likes

- Here are some of the application of IOt….

Industrial application:-

1) **Predictive maintenance :-**

➔ Iot sensor monitor equipment health predicting maintenance needs.

2) **Quality control:-**

➔ Iot sensor monitor production lines , detecting defects.

3) **Supply chain optimization:-**

Iot tracks inventory ,shipping and logistics in real time.

Smart home and cities application:-

1) **Home automation:-**

➔ Iot device control lighting , temperature , security.

2) **Intelligent transportation system:-**

➔ Iot optimize traffic flow, parking , and public transportation.

Healthcare application

1) **Remote patient monitoring :-**

- Iot device track patient condition and health metrics remotely.

2) **Medical device integration:-**

- Iot connect medical device ,enable real time data exchange.

agricultural application

1) **Water founder:-**

- Iot tracks under ground water in the land.

Environmental application

1) **Air quality monitoring:-**

➔ Iot sensor tracks pollution level and predict air quality.

2) Water quality monitoring

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3) **Wildlife conservation:-**

- Iot sensor tracks animal population , habits and behaviour.

➔ These are some example of the many application of iot.

Q-4 Explain Challenges in iot.

Ans:-

- Here are some of the key challenges in IoT:
  - Security challenges
  - Connectivity and communication challenges
  - Data management challenges
  - Scalability challenges
  - Other challenges

1) **Security Challenges:-**

→ **Data Break**: IoT devices collect and transmit sensitive data, making them more secure to avoid data break.

→ **Device risk:** IoT devices often have weak passwords, outdated software, and other risk factor that can be easy take our data by hackers.

3. **Lack of Encryption**: Although encryption is a great way to prevent hackers from accessing data, it is also one of the leading IoT security challenges.

* Many IoT devices do not use encryption, making it easy for hackers to intercept and manipulate data.

2) **Connectivity and Communication Challenges***

1. **Interoperability**: IoT devices from different manufacturers often have different communication protocols, making it difficult for them to communicate with each other.

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2. **Connectivity Issues**: IoT devices require stable and reliable connectivity, which can be a challenge in areas with poor internet connectivity.

3. **Data Transmission**: IoT devices generate large amounts of data, which can be challenging to transmit and process.

3) **Data Management Challenges:-**

→ **Data Volume:** IoT devices generate Large amounts of data, which can be challenging to store, process, and analyze.

→ **Data Quality**: IoT devices can generate inaccurate or incomplete data, which can affect the accuracy of analytics and decision-making.

4) **Scalability Challenges***

-- **Device Management**: As the number of IoT devices increases, managing and maintaining them becomes a one of most challenge.

-- **Scalable Infrastructure:** IoT applications require scalable infrastructure to handle the large amounts of data generated by IoT devices.

-- Scalable Analytics: IoT applications require scalable analytics capabilities to process and analyze the large amounts of data generated by IoT devices.

5) **Other Challenges***

-- **Power Consumption:** IoT devices require power to operate, which can be a challenge in areas where power is limited or unreliable.

-- **Cost**: IoT devices and applications can be expensive to develop.

Q-5. Explain opportunities in iot.

Ans:-

- Key areas where IoT creates opportunities:

- **Industrial Automation:**
  - Monitoring machine health in real-time to predict failures and prevent downtime.
  - Improving production processes by adjusting parameters based on live data.
  - Automated quality control through sensor-based inspections.
- **Smart Cities:**
  - Managing traffic flow with intelligent traffic light systems.
  - Monitoring environmental conditions like air quality.
- **Healthcare:**
  - Remote patient monitoring with wearable devices
  - Real-time tracking of medical supplies and equipment
  - Personalized healthcare insights based on collected data
- **Retail**
  - Customer behavior analysis through in-store sensors
- **Logistics and Supply Chain:**
  - Tracking shipments in real-time with GPS.
  - Optimizing delivery routes based on traffic conditions

Career Opportunities in IoT:

- **Hardware Development:** Designing and developing systems, sensors, and microcontrollers for IoT devices.
- **Software Development:** Creating applications and platforms to manage and analyze IoT data
- **Data Scientist/Analyst:** Extracting meaningful insights from large volumes of IoT data
- **Network Engineer:** Designing and managing secure IoT networks

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- **Cybersecurity Expert:** Protecting IoT devices and systems from cyber threats
- **Solution Architect:** Designing and implementing IoT solutions for specific business needs

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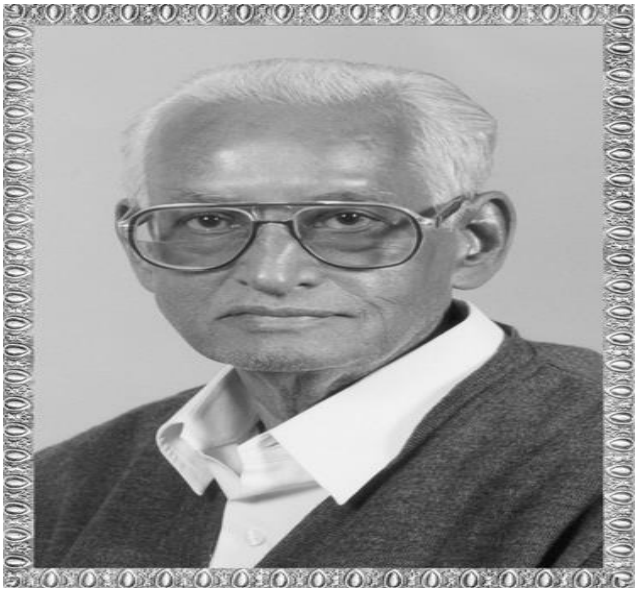

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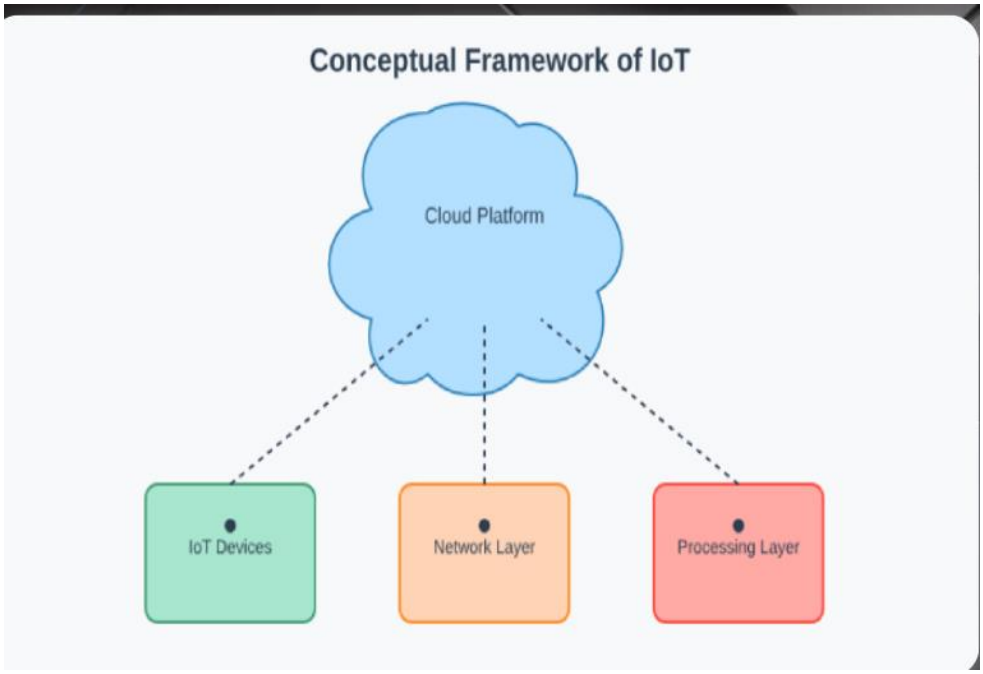
UNIT-2 IOT ARCHITECTURE AND FRAMEWORK

INDEX

TOPICS
*Conceptual Framework • IoT Architecture Overview • Technology behind IoT • Sources of the IoT • M2M Communication • IoT Examples

Q-1. Explain Conceptual Framework of IOT.

Ans:-



- An **IoT framework** can be defined as a set of protocols, tools, and standards that provide a specific structure for developing and manage IoT applications and services.
- In other words, an IoT framework gives you the basics for building your own application.
- The framework of IoT typically includes a combination of the following:

➔ Hardware
➔ Software
➔ Networking elements (IoT protocols)
➔ Device management
➔ Security
➔ Data management

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➔ Application development
➔ Cloud-based platform.

- These components work together to enable theproper integration of IoT devices and systems.
- For example, the device management is necessary for updating and monitoring the performance of the device.
- Protocols allow the different connections between devices and the internet.
- There needs to be a cloud platform where data will be processed and stored, this also connects with an application or platform that is in charge of displaying the data and allows other functions or services.
- **Let us see how the framework can be categorized into four layers-**

1) **Connectivity Layer:**
2) **Access Layer**
3) **Abstraction Layer**
4) **Service Layer.**

1) Connectivity Layer:-

- This layer include all the physical devices included in the framework and the established interconnectivity between them.
- It also includes allocating low-range networking tools like actuators in IoT, sensors in IoT, etc.
- It also enables sharing and distribution of data among numbers of networks.
- It can also share data over a network that is divided into lots of domains.

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2) Access Layer:-

- Data can reaches the internet through the IoT gateway as collected by short range devices in the form of raw information.
- Access layer basically includes network starting, topology definition, and creation of domains, etc.
- This layer also include all(inter or Outer) domain communication, connection setup, scheduling, and packet transmissions between IoT gateway and flow sensors.

3) Abstraction Layer:-

- The core objectives of the abstraction layer is to minimize complexity, ensure faster integration, improve flexibility, and reduce direct dependencies on technologies.
- The key characteristics include management of device-level interactions, translation of protocol languages, providing support to diverse communication standards, and ensuring a uniform communication interface.

4) Service Layer:-

- This is also a imported layer responsible for storing and securing data.
- It also enables data accessibility effectively.
- Its characteristics include integrating information to improve service intelligence, supporting data transformation, providing service discovery, handling data processing, etc.
- Apart from that, it also facilitates communication between devices.
- The core components of the service layer includes resource allocation, data processing services, security management, communication protocols, and service discovery mechanisms.

Q-2 **Explain IOT Architecture in detail.**

Ans:-

- ➔ There are different phases in the architecture of IoT but they can change according to the situations but generally, there are four phases in the architecture of IoT.
- ➔ Which can join physical device ,network and applications to collect ,process and work on data.
- ➔ Layers available in iot architecture are….

1) Device layer
2) Network layer
3) Processing layer
4) Application layer.

1) <u>Device layer:-</u>

- It include all the iot devices such as sensor, devices, actuators and more .
- which collect data from the physical environment, processes it and then sends it over the network.

2) <u>Network Layer:-</u>

- This layer enable communication between device and cloud/server using protocol such wi-fi, blue tooth, and cellular network.
- It convert analog data into digital data.
- It also perform data management and detect problems.

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3) **Processing Layer:-**

- This is very important layer in architecture of iot.
- This layer process and analysis data collected from devices using technology such as cloud computing ,big data analytics and machine learning.

4) **Application layer:-**

- The fourth stage of IoT consists of Cloud/Data Centres where data is managed and used by applications like agriculture, defence, health care etc.

➔ **Components of IOT Architecture:-**

- Iot architecture include many key components likes

1) **Devices:-**

- Sensors, actuators ,controller and other physical devices that collect and transmit data.

2) **Gateway:-**

- It is one device that connect with all other devices.
- It provide protocol conversion ,data processing and security.

3) **Network:-**

- Communication network that enable data transmission between devices and cloud server.

4) **Cloud/Server:-**

- Centralized infrastructure that process ,store, and analysis data.

5) **Database:-**

- Data storage solution that manage and provide access to iot data such as rdbms ,NoSQL database.

➔ **IOT Architecture Pattern:-**

- Many architecture pattern are used in iot system like—

a. **Device to Cloud:-**

➔ Devices sent data directly to the cloud or server.

b. **Device to Gateway:-**

➔ Device sent data to gateway ,which process and forward the data to cloud or server.

c. **Device to Device:-**

➔ Devices communication directly with each other.

Q-3 Explain Technology behind the IOT.

Ans:-

- Iot is a complex system that use multiple technology to enable communication ,data exchange and automation between devices.
- Here are some of the key technology behind IOT

➔ **Hardware components:-**

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1) Microcontrollers:- small computer that control and interact with devices.
2) Sensor:- device that detect and measure parameter like temperature ,humidity and motion.
3) Actuators:- device that perform action based on sensor data.

- **Communication protocol:-**

1) HTTP
2) MQTT(lightweight messaging protocol)
3) Bluetooth low energy.

- **Network infrastructure:-**

1) Wireless sensor network
2) Cellular network
3) Wi-fi network

- **Data Processing and analytics:-**

1) Cloud computing: remote server for data processing , storage and analytics.
2) Edge computing:- local processing of data at the device or gateway
3) machine learning**:-** algorithm for iot data and device behaviour.

- **Security:-**

1) Encryption:- protect data on transfer.
2) Authentication:- verifying device and user identification
3) Access control:- managing device and data access.

- **Operating system:-**

1) RIOT:- real time os for iot device
2) Contiki:-open source os for iot
3) Free RTOS:- real time os for micro controllers.
4) Android things:- a variant of the android os for iot.

Q- Explain Sources of IOT.

Ans:-

- Here are some sources of IOT
- **Hardware Sources:-**

a) **Sensor**: temperature ,pressure and other sensors provide data to iot system.
b) **Micro Controller:**- small computer enable iot device development
c) **System On Chip:**- integrated circuit for iot

- **Software Resources:**-

a) **Operating system:**- iot operating os like freertos, Contiki and android things.
b) **Programming language:**- languages like c, c++,Python and java are use for iot devices programming.
c) **Iot platform:**- platform like aws iot, google cloud, Microsoft platform.
d) **API & SDK:**- application programming interface and software development kit enable iot device integration.

- **Network resources:**-

a) **Wifi**
b) **Bluetooth**

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c) **Cellular network**

- **Energy sources:-**

  a) **Batteries**
  b) **Solar power**
  c) **Wireless power transfer.**

Q- EXPLAIN M2M COMMUNICATION IN DETAIL.

ANS:-

- Machine to machine communication in iot refer to the automated exchange of information between devices, machines or sensors without human intervention.

➔ **Key characteristics of m2m communication:-**

1. **Automated:-** * m2m communication occurs automatically, without intervention of human.
2. **Device to Device:-** * communication take place directly between devices ,machine or sensor.
3. **Wireless:-** m2m communication use wireless technologies , such as cellular network , wifi or blue tooth.

➔ **Application of M2M communication in IOt:**

1. Industrial automation
2. Smart homes
3. Transportation
4. Health care
5. Manufacturing

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6. Utilities

➔ **Types of M2M Communication:-**

- **Cellular-based M2M:** Utilizes cellular networks such as 4G or 5G for communication between devices.
- **Satellite-based M2M:** Utilizes satellite based network.
- **RFID-based M2M:-** used radio frequency identification technology.

➔ **Benefits of M2M Communication in IOT:-**

1. **Increase efficiency:**- automation and real time communication improve productivity and reduce the error compare to human.
2. **Improve safety :-** M2M enable real time monitoring ad alerts , reduce the risk of accident and improving emergency response.
3. **Better services for customer:**- M2M enables personalized services and real time feed back , improving customer satisfaction.
4. **Better supply chain management**

➔ **Challenges & Limitation of M2M:**-

1. **security :-**
2. **communication gap(interoperability)**
3. **standardization**
4. **control device.**

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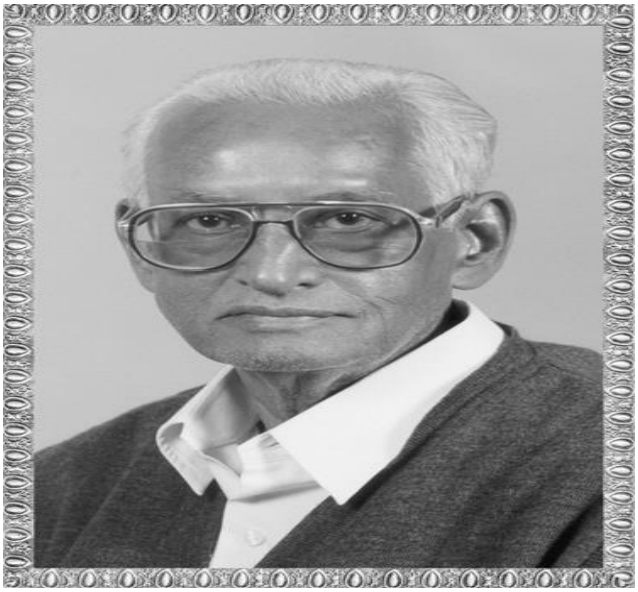

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UNIT-3 HARDWARE FOR IOT

INDEX

TOPICS
• Sensors • Digital Sensors • Actuators • Radio Frequency Identification (RFID) Technology • Wireless sensor networks • Overview of IoT supported Hardware platforms: o Arduino o Netduino

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Q-1 Explain sensor and Digital Sensor.

- Sensor are devices that detect and measure physical parameter such as temperature ,humidity , light and motion.
- They play important role in various application like industrial automation , robotics ,health care and more.

➔ **Types of sensor:-**
- Analog sensor
- Digital sensor

1) Analog sensor:-
- An analog sensor is a device that measures physical quantities like temperature, pressure, light, and sound, and outputs a continuous signal.
- Analog sensors are used in many applications, including industrial control systems, medical devices, and environmental monitoring.
- It collect data using signal or voltage.
- It has low cost.

2) Digital sensor:-
- Digital sensors have been developed to overcome the traditional disadvantages of analog sensors. Digital sensors are widely used in water and industrial processes.
- The first digital sensor was developed in the year 1982.
- Digital sensor directly convert the physical quantity into digital signal.
- This conversion can be easily processed and transmitted by digital system.

- **Advantages:-**

➔ It has more accuracy
➔ Easy to integration with digital system.
➔ Digital sensors are compatible with modern automation systems.
➔

- **Example of digial sensor:-**

1) **Digital thermometer:-**
   - measure temperature and provide digital output.
2) **Optical sensor:-**
   - Detect light and provide a digital output.
3) **Analog to digital conversion:-**
   - Digital signal processed and transmitted to digital system.

- **Additional information**

- Analog sensors are simpler and less expensive than digital sensors.
- Digital sensors are more accurate and easier to integrate than analog sensors.
- Digital sensors are often used in battery-operated systems because they consume less power.
- The resolution of a digital sensor is based on the number of bits used in signal conversion.

Q-2:- Explain actuators in detail.

Ans:-

- An IoT device is made up of a Physical object ("thing") + Controller ("brain") + Sensors + Actuators + Networks (Internet).
- An actuator is a machine component or system that moves or controls the mechanism of the system.
- Sensors in the device sense the environment, then control signals are generated for the actuators according to the actions needed to perform.
- Actuators play important role in iot.
- They are responsible for converting electrical signal into physical actions.
- It allow iot devices to interact with and control the physical world.
- A device that turns electrical energy into mechanical energy is known as an actuator

➔ **Types of actuators:-**

a) Electrical actuators:-

- Convert electrical energy into mechanical energy.

b) Hydraulic actuators:-

c) Gas actuators.

d) Thermal actuators.

➔ **Application of actuators in iot:-**

a) Smart homes:-

- Actuators control lighting, temperature, security system and appliances.

b) Industrial automation:-

- Actuators control machinery ,valves, pumps and belts.

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c) Transportation system:-

- Actuators control traffic light , parking system and autonomous vehicles.

d) Healthcare:-

- It control medical device such as insulin pumps and ventilators.

➔ **Characteristics of Actuators:-**

- **Accuracy**
- **Speed**
- **Reliable**

➔ **Example of actuators:-**

- **Security camera**
- **Wearables**
- **Automatic vehicle**

➔ **Limitation of actuators:-**

- **Communication gap**
- **Power consumption**
- **Maintenance**
- **Cost**

Q-3 Explain RFID Technology.

Ans:-

- Radio frequency identification technology used in wireless communication device.
- Rfid is a wireless communication method used to identify and track object , people or animal using radio waves.
- It use radio waves technology.
- It is one types of chip.
- It is used in sensor.
- RFID is a wireless technology that uses electromagnetic fields to identify and track objects or people.

USe OF RFID:-

1) **Inventory management:-**
   - Tracking product and supplies.
2) **Supply chain management:**
   - Monitoring goods movement and storage.

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3) **Health care:-**
- Tracking patients & medical reports.

4) **Payment system:**
- Contactless payment method, such as credit cards

5) **Fastag:-**
- Used in fastag system in current tall tax system.
- Punch card system used rfid.
- Every office and organization also used RFID to take attendance of their staff.

Benefits Of RFID:-

- **Increase efficiency** :- automating data collection and reduce the error.
- **Imporved Accuracy:-** real time tracking and identification.
- **Reduce the cost:-** minimizing labor cost and improving supply chain management.

Limitation Of RFID:-
- **Cost**
- **Security risk**
- **standard**

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How RFID Works?

1) **RFID TAG:-**

- a small tag or label attached to object , containing a micro chip and antenna.
- RFID tags are made up of an integrated circuit (IC),

2) **RFID READER:-**

- A device that generate radio waves to communicate with the rfid tag.

3) **COMMUNICATION:-**

- The rfid reader send radio waves to the tag, which responds with its stored information.

- **Types of RFID**

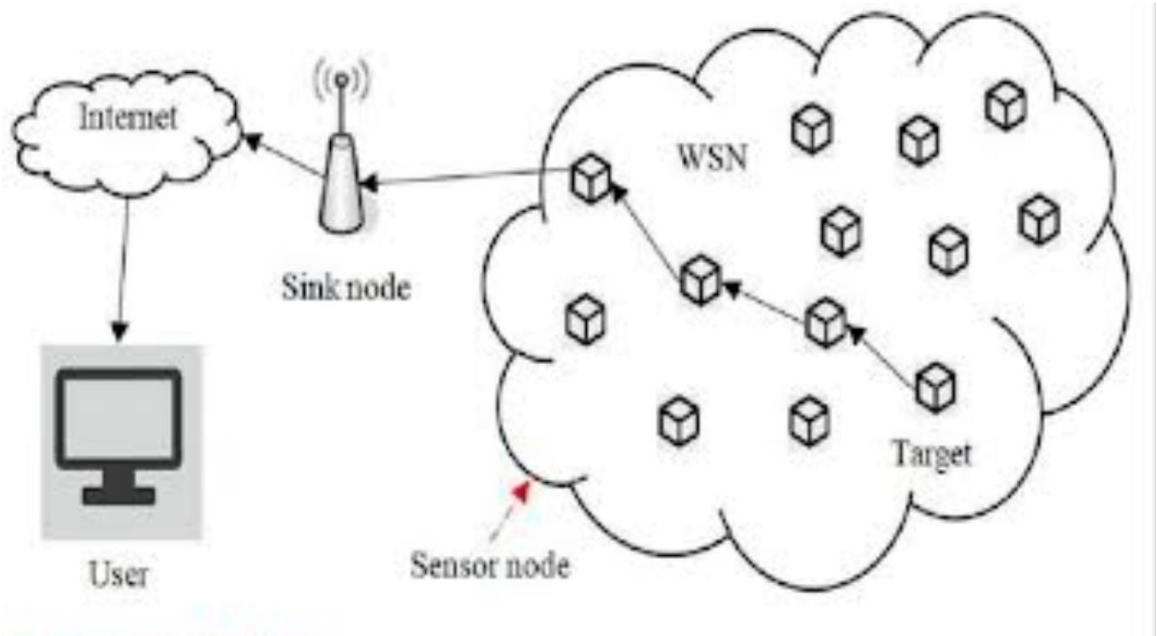
1) **Active RFID:-**

2) **Passive RFID:-**

Q-5 Explain Wireless Sensor Network(WSN)

Ans:-

- A wireless sensor network is a important component of IOT.
- That consist of multiple interconnected sensor node communicating wirelessly.
- These network are used for monitoring and collect data from environment and transmit it for processing and decision making.



- A wsn consist of multiple small sensor node that work together to collect , process and transmit data.

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Components of WSN

1) **Sensor Node:-**

- Small automatic device that sense environmental data like temperature , humidity and motion.
- It collect real world data using sensor.
- Process and store data temporary.
- Communication with neighbouring node.
- It consume low power
- It is in small in size.
- It used wireless communication protocol.

2) **Sink node (Base station):-**

- It is a powerful node that collect data form multiple sensor node and forward it to main network or cloud.
- It is gateway or bridge between sensor node and internet.
- Communication with cloud and user device.

3) **Communication network:-**

- It is one type of medium that allow sensor node to transmit data to sink node.
- It provide short rang like blue tooth or wi fi.
- It also provide long rang like 5g and 6g network.

4) **Power source:-**

- It provide energy to sensor for continue operation.
- Likes battery operand , solar panel or hybrid power.

5) **Data processing:-**

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- Software that processes sensor data before sending it to the cloud or user.

6) **End user interface:-**

- The final component where user interact with WSN data via application.
- Types of interfaces…

➔ Mobile app
➔ Web dash board
➔ Automated control system.

- **Advantages of WSN:-**

➔ **Wireless and easy to use.**
➔ **Low cost**
➔ **Real time monitoring**

- **Limitation of WSN**

➔ **Limited battery**
➔ **Data security**
➔ **Standard.**

Where to use WSN:

- In smart home automation
- Motion direction(light of and on)
- Industrial automation:- machine monitoring.
- Environment monitoring:- air quality , water quality monitor.
- Traffic management
- Waste management(garbage collection monitoring)

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Note:- Not all the IOT technology require the use of WSN. Other technology used instead of WSN like RFID, CELLULAR BASED IOT and many more.

Q-5: Explain hardware platform in IOT.

Or

Explain IOT Supported Hardware platform Overview.

Ans:-

- A platform refer to underlying infrastructure , hardware or software that allow application , program or system to run.
- In computer system there are many platform are available like hardware platform like cpu, ram, mother board and many more., software platform like os , programming language and many more.
- The Internet of Things (IoT) connects a wide variety of devices, sensors, and actuators to collect and exchange data.
- IoT-supported hardware platforms provide the underlying infrastructure necessary for building IoT systems.
- These platforms include microcontrollers, microprocessors, connectivity modules, sensors, and other components that help in the deployment of IoT applications.
- There are two types of platform available in IOT like **1) ARDUINO 2) NETDUINO.**

1) ARDUINO:-

- Arduino is an open-source platform.
- It is used for building digital devices and interactive objects.
- It is based on an easy-to- use hardware and software system that allows users to develop systems for various applications.

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- Arduino is both hardware and software platform .
- A hardware platform: An open-source micro controller board used to create electronic projects.(**it is use micro controller**).
- A software platform: The Arduino Integrated Development Environment (IDE) used to write and upload code to the hardware.
- It is use in back ground of application or IOT device.

➔ **Key Features of Arduino:-**

- Easy to Use: Simplified programming with C/C++.
- Low Cost
- Open-Source: Free to modify and customize.
- Wide Community Support: Large developer base with extensive libraries.
- IoT and Automation: Supports communication modules like WiFi, Bluetooth.

➔ **Arduino Hardware**

1) **Microcontroller:-** The heart of Arduino that processes input signal sand controls output devices.
   - Common micro controllers used:

➔ ATmega328P
➔ ATmega2560

2) **Input/Output Pins:-**

➔ Digital pin: used for high and low signal
➔ Analog pin:- used for sensor.

3) **Power Supply:-**

➔ USB Power(5V):
➔ Pin(7-12V)

➔ Arduino Software:-

1) Arduino IDE:-

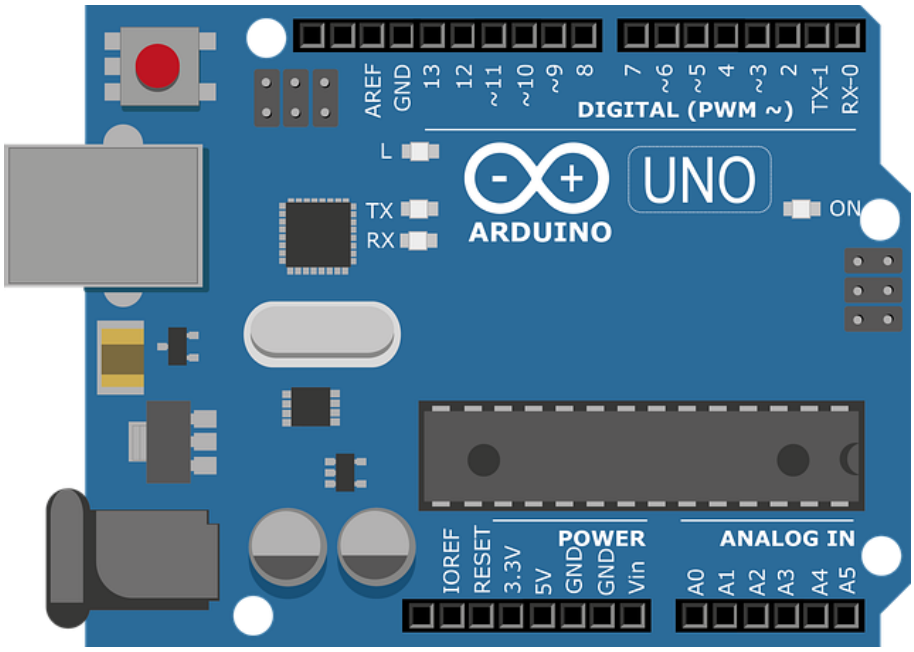
- Open-source software used for writing, compiling, and uploading code.
- Supports C/C++ programming with prebuilt functions.

2) Libraries:-

- Provide in built library for application.

→ Advantages:-

- Easy to use
- Develop for new developer.
- Simple programming model.



➔ NETDUINO.

- Netduino provide both hardware and software platform.
- Netduino is advance and powerful platform.
- Hardware platform:- series of micro controller board that run on .net framework.
- Software platform:- it provide programming environment that allow to write code in c#.
- Compare to Arduino netduino used high level programming language like c#.
- Netduino has powerful processor like cortex-m3 which is more powerful that Arduino processor.
- It provide 32 bit processor.
- Allow for easy communication with computer and other device.
- Netduino is also open source but it has more powerful than Arduino.
- Netduino is integrated with visual studio.

➔ Netduino hardware:-

1) **Micro controller**:- arm based micro controller. (it is more powerful then Arduino processor)
2) **Input pins:-** it also used digital and ana log pin.
3) **Power supply:-** usb power and pin power.

➔ Netduino software:-

1) **Development environment:-**

➔ Uses Visual Studio instead of the Arduino IDE. • Supports C# and .NET Micro Framework, making it easy for developers in the .NET ecosystem.

2) **Libraries:-**

➔ Microsoft's.NET Framework Libraries simplify development.

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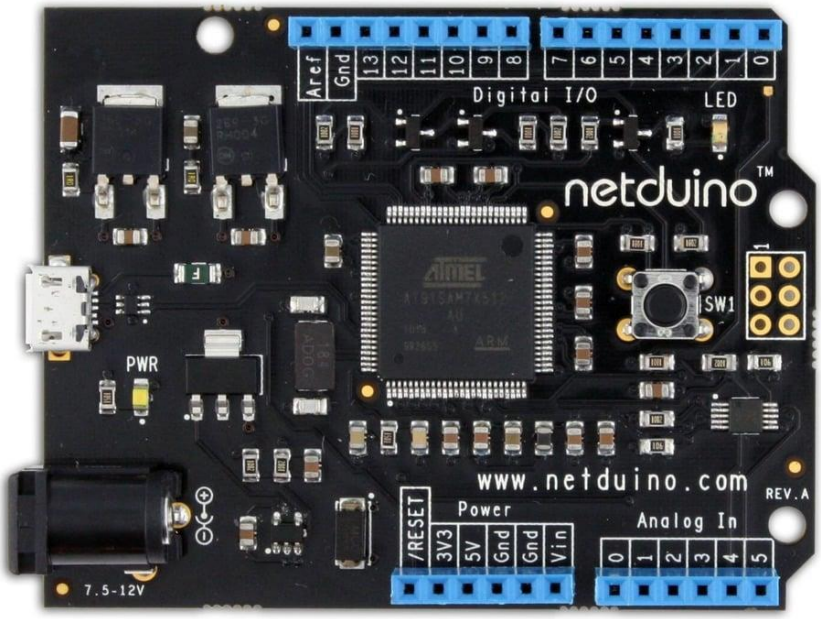
➔ Supports cloud-based services and web APIs, making it more advanced than Arduino for IoT projects.

- Netduino is a powerful platform for system development, offering the benefits of C# programming and .NET integration.
- It is well- suited for IoT applications, providing network connectivity and easy integration with sensors and actuators.